Shreyas G Shet

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Education

BMS Institute of Technology & Management Bengaluru, IN

Bachelor of Engineering, Computer Science and Engineering *2023 — 2027*

- CGPA: 8.72
- Relevant Coursework: Data Structures and Algorithms, Operating Systems, Digital Design and Computer Organization, Database Management Systems, Theory of Computation, Compiler Design, Computer Networks, Cloud Computing, Machine Learning, Microprocessors, Linear Algebra, Discrete Mathematics, Multivariable & Single Variable Calculus

Projects

Custom Memory Allocator GitHub

Tech: C, Linux/UNIX, Makefile, GDB

- Designed and implemented a heap allocator in C from scratch, replicating core mechanics of production allocators like glibc's malloc: explicit free lists, first-fit search, block splitting, forward/backward coalescing, and 16-byte aligned allocation with boundary-tag headers.
- Modeled sbrk-style heap growth on a simulated process heap to mirror real UNIX memory management, so the allocator's logic maps directly onto how an OS actually grows and reclaims a process's address space.
- Achieved 91.2% peak heap utilization and 28.6% free-block reuse under controlled workload testing, by instrumenting the allocator to track allocation attempts, splits, and coalescing events, then tuning the splitting/coalescing logic against that data.

Production Task Manager API GitHub

Tech: Java, Spring Boot, REST APIs, Spring Data JPA, Hibernate, PostgreSQL, Maven

- Built a REST API in Java (Spring Boot, Maven, PostgreSQL) supporting CRUD, filtering, and status workflows for tasks.
- Centralized error handling with `@ControllerAdvice`, returning consistent 400/404/500 responses instead of per-endpoint status-code logic.
- Replaced file-based JSON storage with PostgreSQL (JPA/Hibernate) after stress testing confirmed a race condition and exposed the limits of file storage, with NOT NULL and CHECK constraints enforced at the schema level.

Mini Inference Engine GitHub

Tech: Python, PyTorch, NumPy, HuggingFace Transformers, GPT-2

- Implemented the GPT-2 forward pass from scratch as plain functions (layer norm, causal multi-head attention, MLP, pre-LN transformer block), on NumPy arrays wihtout nn.Module classes or library model code.
- Loaded pretrained GPT-2 weights, including token and position embeddings, from the HuggingFace checkpoint, and used PyTorch and NumPy for tensor math and weight handling.
- Verified correctness against HuggingFace's reference GPT-2: next-token argmax matched at every position across 4 test prompts, with a max absolute logit difference of 1.2e-4.
- Extending the engine with a KV cache and continuous batching, to be benchmarked against static batching on tokens/sec and memory utilization.

Learning-Based Compiler Optimization Pass Selection using Agentic Systems GitHub

Tech: LLVM, LLVM IR, C, Python

- Designed a first-principles methodology for compiler pass-ordering research: identified optimization opportunities in LLVM IR by hand, mapped them to specific LLVM passes, and empirically tested sequencing effects across a 6-pass pipeline (mem2reg, instcombine, simplifycfg, reassociate, newgvn, gvn-hoist).
- Laid the technical foundation for an agentic compiler-optimization system that observes program IR state, selects/applies LLVM passes, and adapts based on newly exposed opportunities — motivated by the research question of whether learned pass sequences can beat static -O2/-O3 pipelines.

Skills

- **Programming Languages**: C, C++, Java, Python, Bash, SQL
- **Backend and Databases**: Spring Boot, REST APIs, PostgreSQL, JPA, Hibernate, Maven
- **Tooling**: Linux, UNIX, Git, GitHub Actions, Continuous Integration/Continuous Delivery (CI/CD), Docker, NGINX
- **Machine Learning**: PyTorch, HuggingFace Transformers, NumPy, GPT-2, Large Language Model (LLM) inference